Physics 9702 Paper 4 · Topic 21

Alternating currents

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/41 October/November 2023, Question 7 (8 marks) page 2

Examiner report: (b) Candidates found some aspects of this sketch difficult.

2 9702/41 October/November 2022, Question 7 (10 marks) page 4

Examiner report: (ii) The most common misconception here was with the shape of the line drawn. The fact the line was symmetrical about this value was not often known, and candidates who had drawn an incorrect shape of line without the relevant symmetry found it difficult to use their line to answer this question.

3 9702/42 May/June 2023, Question 7 (10 marks) page 6

Examiner report: (iii) Candidates found this a very challenging question. Whilst many candidates did correctly conclude that the r.m.s. voltages are equal, most found it difficult to adequately articulate the reason why this is the case.

4 9702/41 May/June 2025, Question 6 (12 marks) page 8

Examiner report: (iii) Most candidates had the correct symbols for four diodes, but only the strongest candidates drew a circuit diagram which would work as a full-wave rectifier, with all of the diodes in the correct orientation.

5 9702/44 October/November 2025, Question 6 (15 marks) page 11

Examiner report: A significant minority of candidates had incorrect orientations of one or more of the diodes.

6 9702/41 May/June 2024, Question 7 (10 marks) page 14

Examiner report: (ii) The majority of the candidates found it difficult to appreciate the impact of squaring the voltage to produce the sinusoidal shape of the power–time graph, and they repeated a full-wave rectification graph instead.

7 9702/41 May/June 2022, Question 5 (12 marks) page 16

Examiner report: (iii) This calculation proved challenging.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

7 A varying current I passes through a resistor of resistance R in the circuit shown in Fig. 7.1.

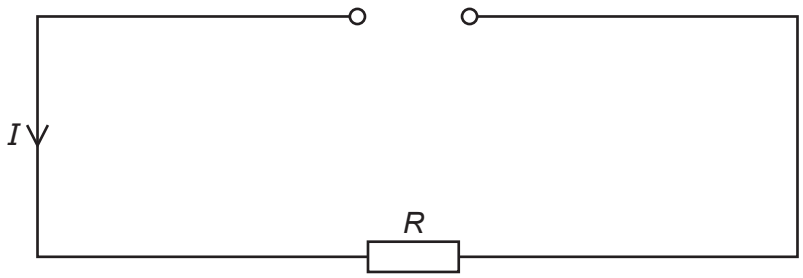


Fig. 7.1

Fig. 7.2 shows the variation with time t of I.

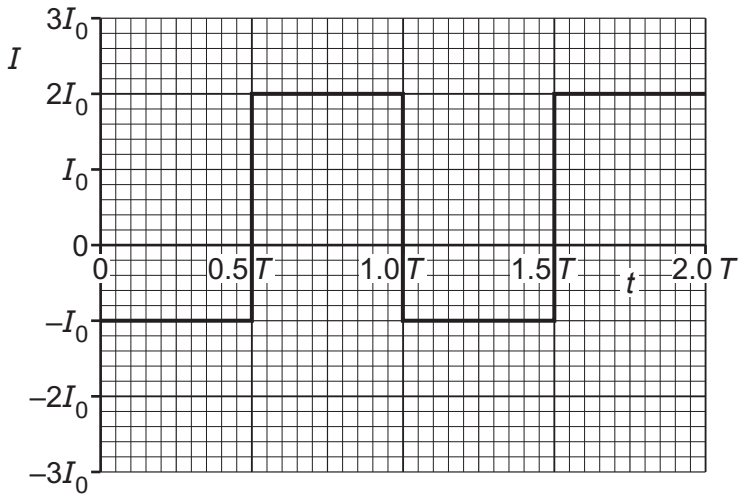


Fig. 7.2

The current has magnitude $2I_0$ when it is in the positive direction and I_0 when it is in the negative direction. The period of the variation of the current is T.

(a) Determine expressions, in terms of I_0 and R, for the power P dissipated in the resistor for the times when:

(i) the current is in the negative direction

P = ... [1]

(ii) the current is in the positive direction.

P = ... [1]

(b) On Fig. 7.3, sketch the variation of P with t between $t = 0$ and $t = 2.0T$. Label the power axis with an appropriate scale.

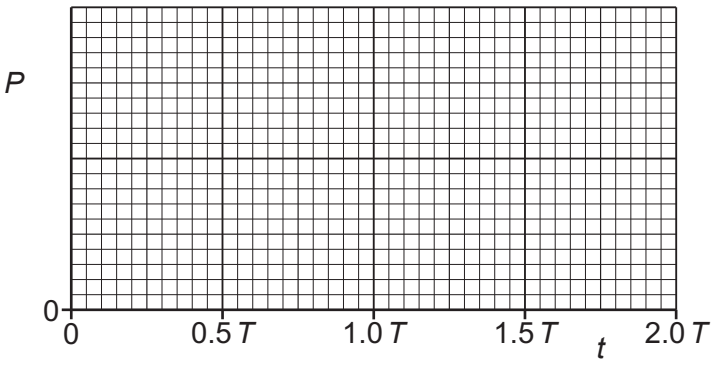


Fig. 7.3

[3]

(c) Use your answer in **(b)** to determine an expression, in terms of I_0 and R, for:

(i) the mean power $\langle P \rangle$ in the resistor

$\langle P \rangle$ = ... [1]

(ii) the root-mean-square (r.m.s.) current $I_{\text{r.m.s.}}$ in the resistor.

$I_{\text{r.m.s.}}$ = ... [2]

[Total: 8]

7 **(a)** A sinusoidal alternating voltage has a root-mean-square (r.m.s.) potential difference (p.d.) of 4.2 V and a frequency of 50 kHz.

(i) The alternating voltage is applied across a resistor of resistance 760 Ω.

By considering the peak voltage, show that the maximum power dissipated by the resistor is 46 mW.

[2]

(ii) On Fig. 7.1, draw a smooth curve to show how the power P dissipated in the resistor varies with time t between $t = 0$ and $t = 40\,\mu s$. Assume that $P = 0$ when $t = 0$.

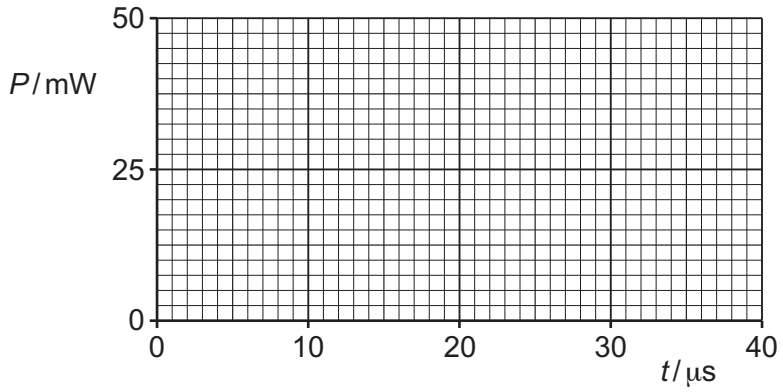


Fig. 7.1

[3]

(iii) Use your line in **(a)(ii)** to explain why the mean power dissipated in the resistor is 23 mW.

.....

.....

..... [1]

(b) The alternating voltage in **(a)** is now applied to a piezoelectric crystal in air.

(i) Explain what happens to the air surrounding the crystal.

.....

.....

.....

..... [3]

(ii) A second piezoelectric crystal is placed in the air near to the first crystal.

Explain the effect of the surrounding air in **(b)(i)** on the second crystal.

..... [1]

[Total: 10]

7 Four diodes are used in a bridge rectifier circuit to produce rectification of a sinusoidal a.c. input voltage V_{IN}.
Fig. 7.1 shows part of the circuit, but three of the diodes are missing.

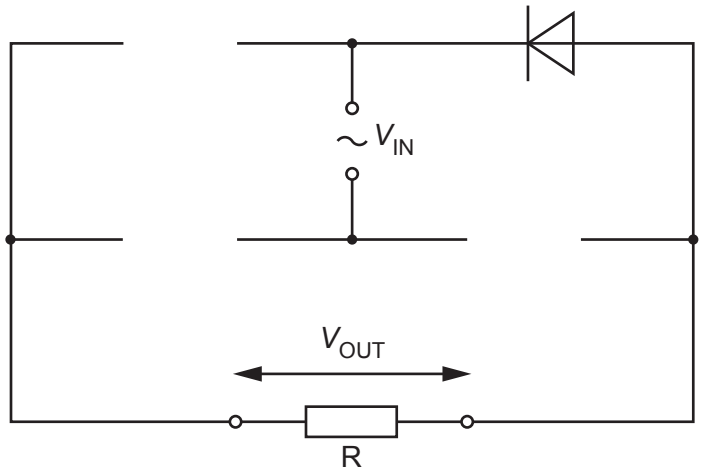


Fig. 7.1

The p.d. across the load resistor R is the output p.d. V_{OUT} of the bridge rectifier.

(a) (i) State the name of the type of rectification produced by a bridge rectifier.

..... [1]

(ii) Complete Fig. 7.1 by drawing the three missing diodes, correctly connected. [2]

(iii) On Fig. 7.1, draw an arrow to indicate the direction of the current in resistor R. [1]

(b) V_{IN} has amplitude V_0 and period T. Fig. 7.2 shows the variation with time t of V_{IN}.

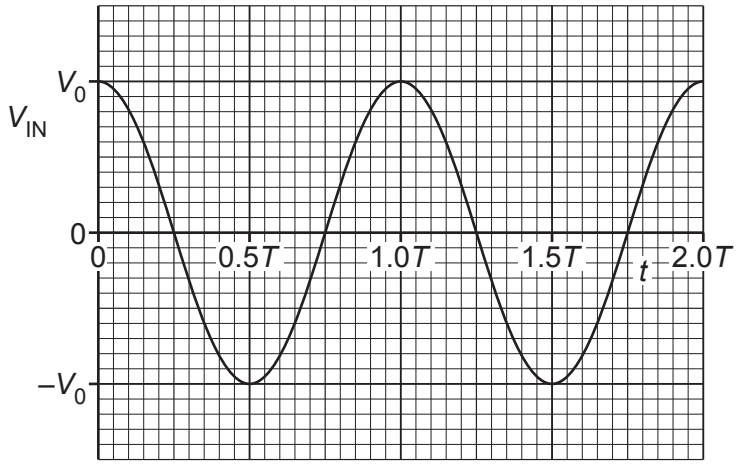


Fig. 7.2

(i) On Fig. 7.3, sketch the variation of V_{OUT} with t between $t = 0$ and $t = 2.0T$.

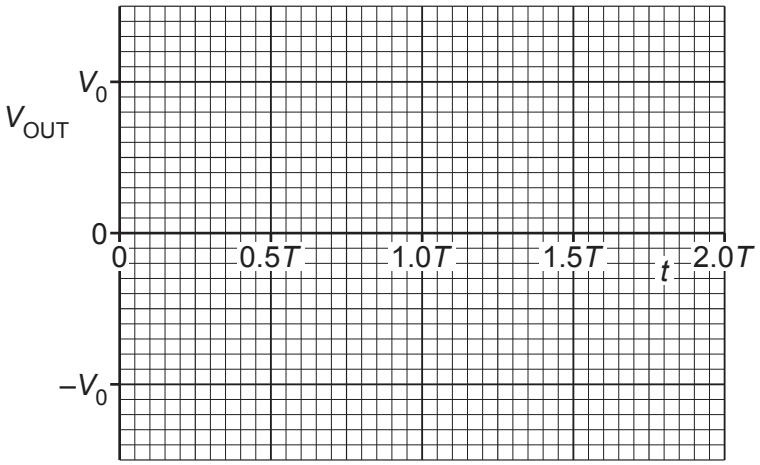


Fig. 7.3

[3]

(ii) The power dissipated in the resistor is P.

On Fig. 7.4, sketch the variation of P with t between $t = 0$ and $t = 2.0T$.

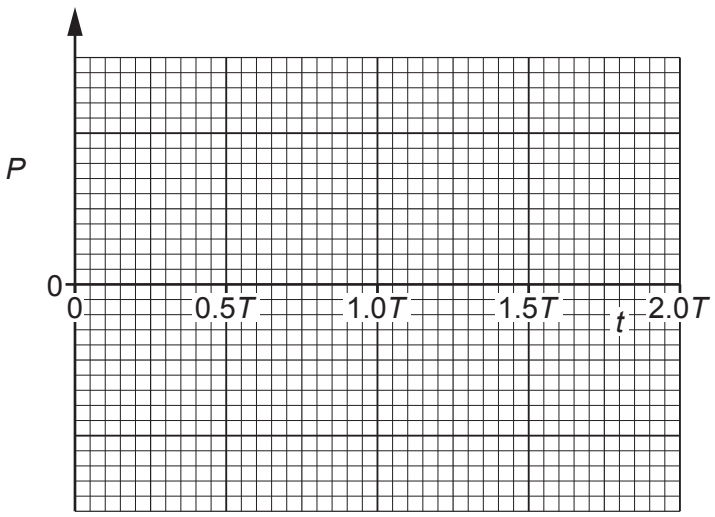


Fig. 7.4

[2]

(iii) Suggest, with a reason, how the root-mean-square (r.m.s.) value of V_{OUT} compares with the r.m.s. value of V_{IN}.

.....

..... [1]

[Total: 10]

6 Fig. 6.1 shows a circuit that rectifies an alternating input voltage V_{IN} and produces an output voltage V_{OUT} across a resistor R.

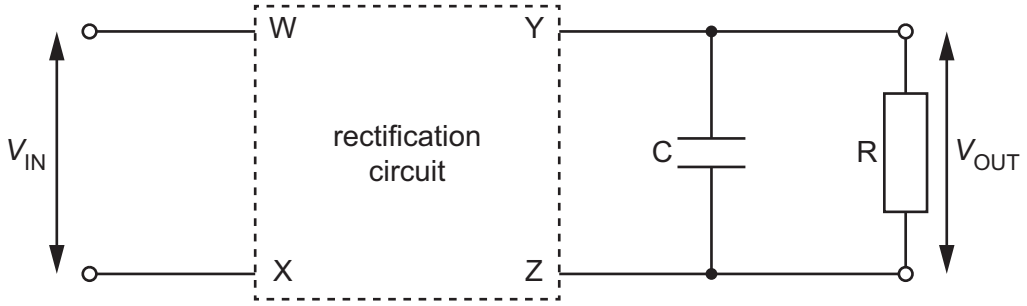


Fig. 6.1

The four terminals of the rectification circuit are labelled W, X, Y and Z.
A capacitor C is connected in parallel with resistor R.

(a) (i) State what is meant by rectification.

.....

..... [1]

(ii) State the purpose of capacitor C.

.....

..... [1]

(b) Fig. 6.2 shows the variations with time t of the potential differences (p.d.s) V_{IN} and V_{OUT}.

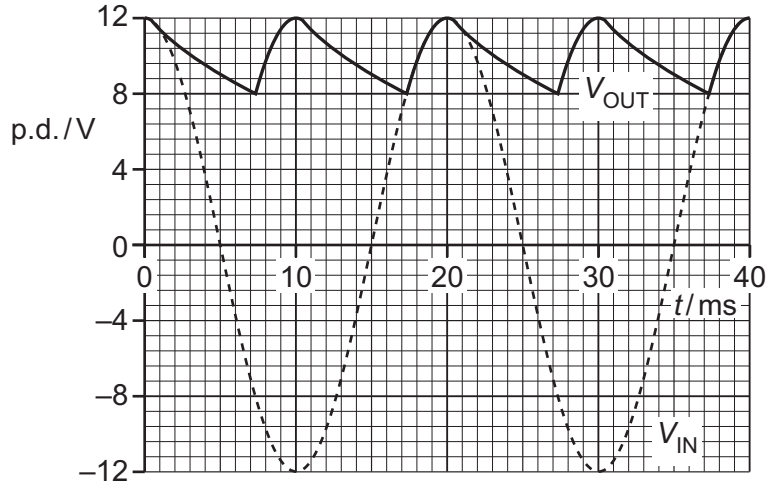


Fig. 6.2

(i) The variation of V_{IN} with t can be represented by

$$V_{IN} = A \cos Bt$$

where A and B are constants.

Determine the values of A and B. Give a unit with your answer for A.

A = ... unit .....

B = ... rad s^{-1} [2]

(ii) Determine the type of rectification produced by the circuit in Fig. 6.1.

..... [1]

(iii) On Fig. 6.3, draw the circuit diagram for the components inside the rectification circuit.

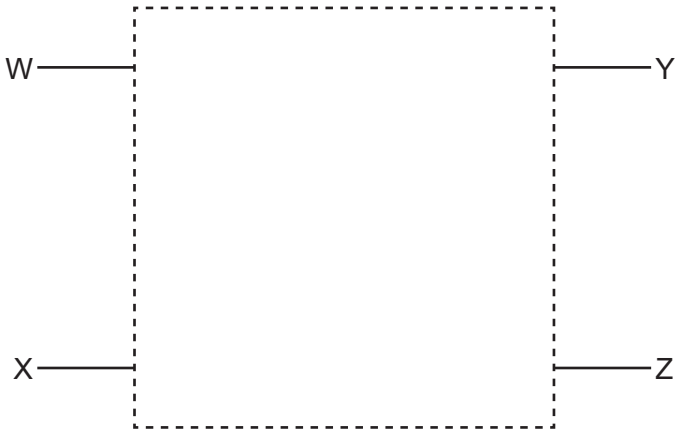


Fig. 6.3

[2]

(iv) Determine a value for the time constant for the discharge of the capacitor C through the resistor R in Fig. 6.1.

time constant = ... s [3]

(c) The capacitor C has a capacitance of 570 μF.

Use your answer in **(b)(iv)** to determine the resistance of resistor R.

resistance = ... Ω [2]

[Total: 12]

6 Fig. 6.1 shows part of a bridge rectifier circuit that can be used for rectification of an alternating input voltage V_{IN}.

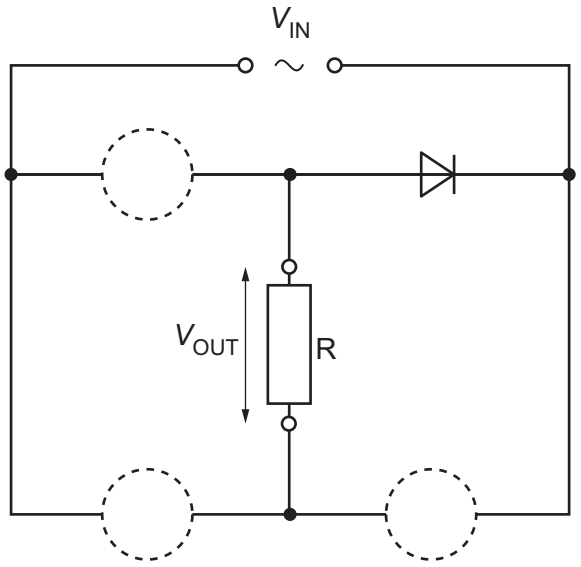


Fig. 6.1

The circuit contains four diodes, one of which is shown.
The rectified output voltage V_{OUT} is applied across load resistor R.

(a) (i) State what is meant by rectification.

.....

..... [1]

(ii) State the name of the type of rectification produced by a bridge rectifier circuit.

..... [1]

(iii) Complete the circuit in Fig. 6.1 by drawing the three missing diodes inside the dashed circles. [2]

(b) The input voltage varies with time t according to the equation

$$V_{IN} = 34 \sin 18t$$

where V_{IN} is in V and t is in s.

(i) Show that the period of the input voltage is 0.35 s.

[2]

(ii) Calculate the root-mean-square (r.m.s.) input voltage.

r.m.s. voltage = ... V [1]

(iii) On Fig. 6.2, sketch the variation of V_{OUT} with t from $t = 0$ to $t = 0.35$ s.

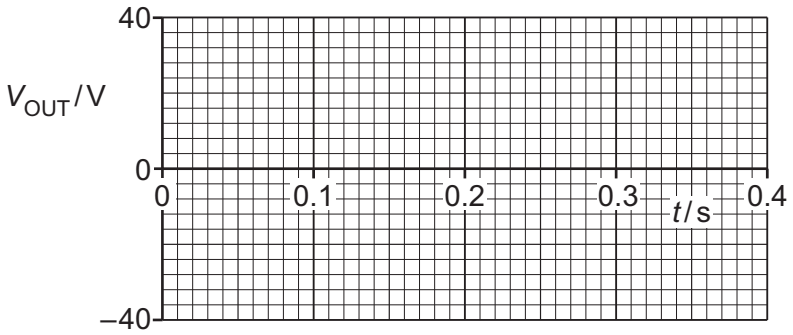


Fig. 6.2

[3]

(c) Resistor R has a resistance of 56 kΩ. A capacitor of capacitance 12 μF is connected into the circuit of Fig. 6.1 in order to smooth the output voltage.

(i) On Fig. 6.1, draw the capacitor correctly connected into the circuit. [1]

(ii) Calculate the time constant of the smoothing circuit.

time constant = ... s [2]

(iii) During each discharge cycle, the time for which the capacitor is discharging is 0.14 s.

Determine the minimum value of the smoothed output voltage.

minimum voltage = ... V [2]

[Total: 15]

7 A circuit contains a power supply that provides a sinusoidal alternating input voltage V_{IN}. There is an output voltage V_{OUT} across a load resistor R, as shown in Fig. 7.1.

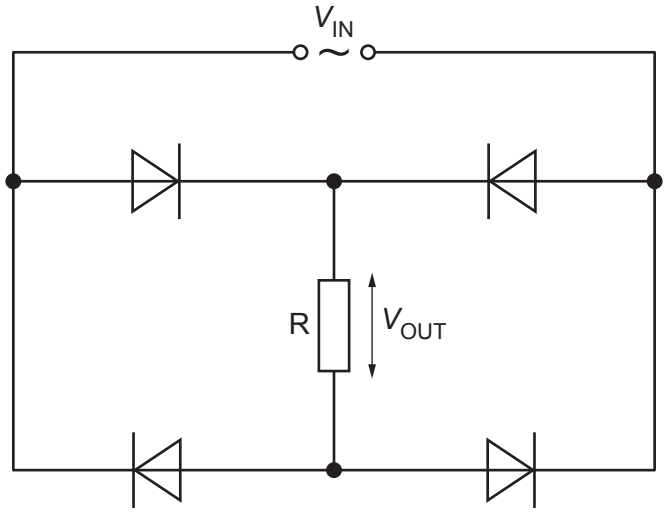


Fig. 7.1

(a) State the purpose of the circuit in Fig. 7.1.

.....

.....

..... [2]

(b) Fig. 7.2 shows the variation of V_{OUT} with time t.

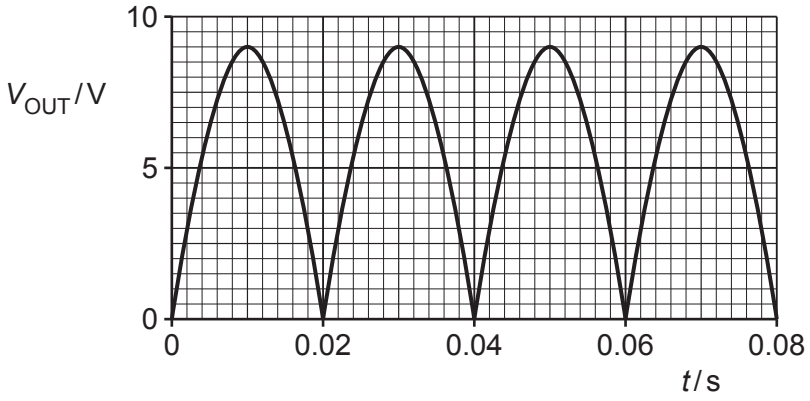


Fig. 7.2

(i) The load resistor R has a resistance of 370 Ω.

Show that the maximum power dissipated in R is 0.22 W.

[2]

(ii) On Fig. 7.3, sketch the variation with *t* of the power *P* dissipated in R.

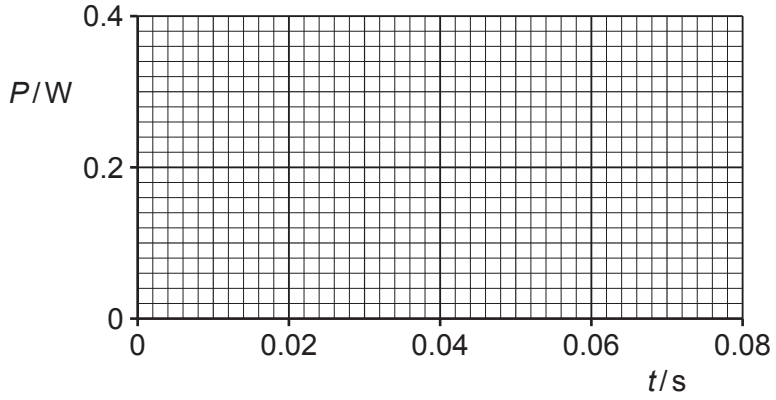


Fig. 7.3

[3]

(iii) Calculate the mean power dissipated in R.

mean power = ... W [1]

(c) The circuit of Fig. 7.1 is disconnected, and R is connected directly across the power supply.

Explain, without calculation, how the mean power now dissipated in R compares with the answer in **(b)(iii)**.

.....

.....

..... [2]

[Total: 10]

5 Fig. 5.1 shows four diodes and a load resistor of resistance 1.2 kΩ, connected in a circuit that is used to produce rectification of an alternating voltage.

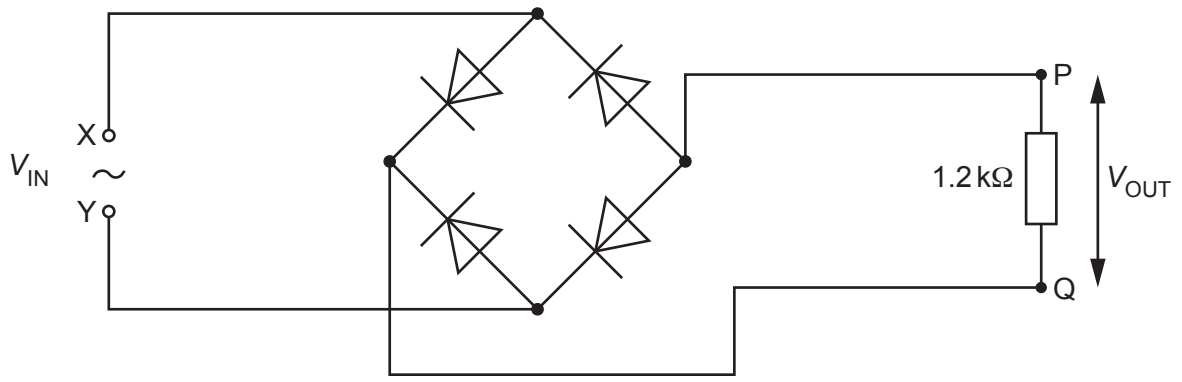


Fig. 5.1

(a) (i) State what is meant by rectification.

.....

..... [1]

(ii) State the type of rectification produced by the circuit in Fig. 5.1.

..... [1]

(b) A sinusoidal alternating voltage V_{IN} is applied across the input terminals X and Y. The variation with time t of V_{IN} is given by the equation

$$V_{IN} = 6.0 \sin 25\pi t$$

where V_{IN} is in volts and t is in seconds.

(i) On Fig. 5.1, label the output terminals P and Q with the appropriate symbols to indicate the polarity of the output voltage V_{OUT}. [1]

(ii) The magnitude of the output voltage V_{OUT} varies with t as shown in Fig. 5.2.

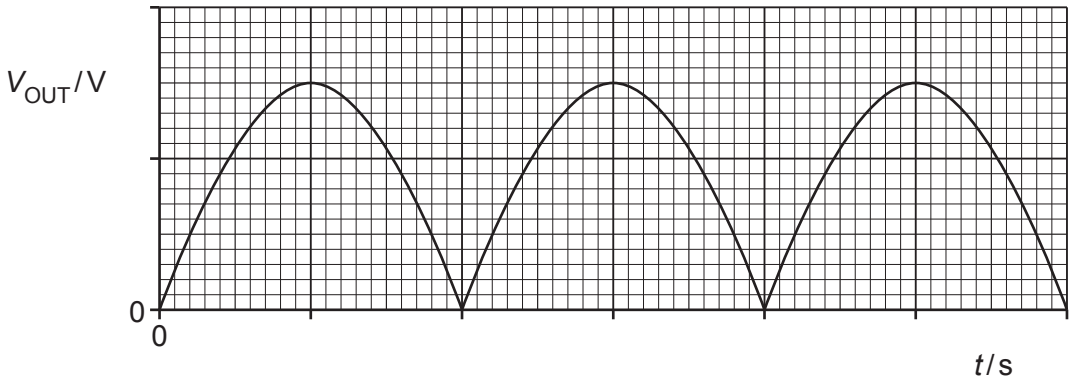


Fig. 5.2

On Fig. 5.2, label both of the axes with the correct scales. Use the space below for any working that you need.

[3]

(c) The output voltage in **(b)** is smoothed by adding a capacitor to the circuit in Fig. 5.1.
The difference between the maximum and minimum values of the smoothed output voltage is 10% of the peak voltage.

(i) On Fig. 5.1, draw the circuit symbol for a capacitor showing the capacitor correctly connected into the circuit. [1]

(ii) On Fig. 5.2, sketch the variation with t of the smoothed output voltage. [2]

(iii) Calculate the capacitance C of the capacitor.

C = ... F [3]

[Total: 12]

Question	Answer	Marks
7(a)(i)	$P = I_0^2R$	**A1**
7(a)(ii)	$P = 4I_0^2R$	**A1**
7(b)	sketch: square wave of period T, with P always non-zero	**B1**
	horizontal lines, from 0 to 0.5T and from 1.0T to 1.5T, all at the same level that the scale indicates to be I_0^2R	**B1**
	horizontal lines, from 0.5T to 1.0T and from 1.5T to 2.0T, at a level that is four times higher than the lower lines	**B1**
7(c)(i)	$\langle P \rangle = (5/2)I_0^2R$	**A1**
7(c)(ii)	$\langle P \rangle = I_{r.m.s.}^2R$	**C1**
	$I_{r.m.s.}^2R = (5/2)I_0^2R$ $I_{r.m.s.} = \sqrt{(5/2)}\ I_0$	**A1**

Question	Answer	Marks
7(a)(i)	peak voltage = $4.2 \times \sqrt{2}$ (= 5.9 V)	**B1**
	power = V^2 / R = $5.9^2 / 760$ = 0.046 W or 46 mW	**A1**
7(a)(ii)	sketch shows peak(s) in power at 46 mW	**B1**
	correct shape (sinusoidal wave sitting on *t*-axis)	**B1**
	four cycles of repeating pattern shown, with P = 0 at 0, 10, 20, 30, 40 μs	**B1**
7(a)(iii)	line is symmetrical about 23 mW	**B1**
7(b)(i)	(alternating p.d. makes) the crystal vibrate	**B1**
	vibrations (of crystal) causes air to vibrate	**B1**
	frequency is in ultrasound range	**B1**
7(b)(ii)	(air makes) crystal vibrate, which causes an e.m.f. to be generated across the (second) crystal	**B1**

Question	Answer	Marks
7(a)(i)	full-wave (rectification)	**B1**
7(a)(ii)	lower left diode shown pointing left	**B1**
	lower right and upper left diodes shown pointing left	**B1**
7(a)(iii)	arrow indicating current direction in resistor to the right	**B1**
7(b)(i)	sketch: periodic line showing minimum $V_{OUT} = 0$ and maximum $V_{OUT} = +V_0$	**B1**
	line showing peak V_{OUT} at $t = 0$, $0.5T$, $1.0T$, $1.5T$ and $2.0T$, with V_{OUT} going to zero half-way in between each peak	**B1**
	line showing correct modulated sine shape	**B1**
7(b)(ii)	sketch: sinusoidal curve with troughs sitting on the time axis	**B1**
	peak power at $t = 0$, $0.5T$, $1.0T$, $1.5T$ and $2.0T$ and zero power half-way in between each peak	**B1**
7(b)(iii)	same power-time graph with or without rectification, so same V_{rms} **or** V^2-time graph is same for both V_{OUT} and V_{IN}, so same V_{rms} **or** power does not depend on sign of V, so same V_{rms}	**B1**

Question	Answer	Marks
6(a)(i)	conversion from a.c. to d.c.	**B1**
6(a)(ii)	smoothing	**B1**
6(b)(i)	$A = 12\ \text{V}$	**A1**
	$B = 2\pi / (20 \times 10^{-3})$ $= 310\ \text{rad s}^{-1}$	**A1**
6(b)(ii)	full-wave (rectification)	**B1**
6(b)(iii)	four diodes shown, with correct circuit symbols	**B1**
	four diodes correctly connected to form a bridge rectifier	**B1**
6(b)(iv)	$V = V_0 \exp(-t / \tau)$ **or** $V = V_0 \exp(-t / RC)$ **and** $\tau = RC$	**C1**
	$8.0 = 12 \exp(-7.3 \times 10^{-3} / \tau)$	**C1**
	$\tau = 0.018\ \text{s}$	**A1**
6(c)	time constant = RC	**C1**
	$R = (0.018 / 570 \times 10^{-6})$ $= 32\ \Omega$	**A1**

Question	Answer	Marks
6(a)(i)	conversion of a.c. to d.c.	**B1**
6(a)(ii)	full-wave (rectification)	**B1**
6(a)(iii)	three diodes with correct symbols connected into the circuit	**B1**
	lower-left diode shown pointing towards the right **and** upper-left and lower-right diodes both shown pointing towards the left	**B1**
6(b)(i)	$\omega = 18$ (rad s^{-1})	**C1**
	$T = 2\pi / 18 = 0.35$ s	**A1**

Question	Answer	Marks
6(b)(ii)	$V_{r.m.s.} = 34 / \sqrt{2}$ $= 24$ V	**A1**
6(b)(iii)	sinusoidal 'humps' with minimum $V_{OUT} = 0$ and non-zero V_{OUT} all same sign	**B1**
	peak V_{OUT} shown as 34 V **or** –34 V	**B1**
	two 'humps' shown, with V_{OUT} always with same sign **and** $V_{OUT} = 0$ at $t = 0$, $t = 0.175$ s and $t = 0.350$ s **and** non-zero in between	**B1**
6(c)(i)	correct circuit symbol for capacitor, connected in parallel with resistor	**B1**
6(c)(ii)	time constant = RC	**C1**
	$= 56 \times 10^3 \times 12 \times 10^{-6}$ $= 0.67$ s	**A1**
6(c)(iii)	$V_{min} = 34 \exp(-0.14 / 0.67)$	**C1**
	$= 28$ V	**A1**

Question	Answer	Marks
7(a)	rectification (of the input voltage)	**M1**
	full-wave	**A1**
7(b)(i)	$P = V^2 / R$ **or** maximum $V = 9.0\,\text{V}$	**C1**
	$P_{MAX} = 9.0^2 / 370 = 0.22\,\text{W}$	**A1**
7(b)(ii)	sinusoidal shape with minima sitting on the time axis	**B1**
	correct frequency and phase, with minima at 0, 0.02, 0.04, 0.06 and 0.08 s and maxima at 0.01, 0.03, 0.05 and 0.07 s	**B1**
	all maxima shown at 0.22 W	**B1**
7(b)(iii)	mean power = peak power / 2 = 0.22 / 2 = 0.11 W	**A1**
7(c)	power–time graph is identical	**B1**
	(so) mean powers are equal	**B1**

Question	Answer	Marks
5(a)(i)	conversion (from a.c.) to d.c.	**B1**
5(a)(ii)	full-wave (rectification)	**B1**
5(b)(i)	P labelled – **and** Q labelled +	**B1**
5(b)(ii)	V_{OUT} scale labelled 4 and 8 on the 2 cm tick marks	**B1**
	$T = 2\pi / \omega$ $= 2\pi / 25\pi$ $= 0.08$ s	**C1**
	t scale labelled 0.02, 0.04, 0.06, 0.08, 0.10, 0.12 on the 2 cm tick marks	**A1**
5(c)(i)	correct symbol used for capacitor and capacitor connected in parallel with the 1.2 kΩ resistor.	**B1**
5(c)(ii)	straight lines or curves, with negative decreasing gradients, drawn between adjacent peaks, from top of first peak to meet line going up to next peak	**B1**
	lines, from one peak to the line going up to the next peak, show a drop in p.d. of 1½ small squares	**B1**
5(c)(iii)	$V = 0.90 \times 6.0$ (= 5.4 V) **or** discharge time (for each cycle) = 0.034 s	**C1**
	$V = V_0 \exp(-t / RC)$ $5.4 = 6.0 \exp[-0.034 / (1.2 \times 10^3 \times C)]$	**C1**
	$C = 2.7 \times 10^{-4}$ F	**A1**